# Lab Four – Programming in R

- Complete the tasks below. Make sure to start your solutions in on a new line that starts with "**Solution**:".
- Make sure to use the Quarto Cheatsheet. This will make completing and writing up the lab *much* easier.

In this lab, we will build a mealkit recipe generator. I created a small website of about 40 recipes. We will scrape recipes from that website, randomly select three meals, and print a grocery list with recipe cards.

## 1 Preliminaries

### 1.1 Part a

Below, I create a numeric vector filled with random observations. Ask for the 3rd item. **Note**: The `set.seed(7272)` portion ensures we all get the same answer.

```
set.seed(7272) # sets the randomization seed for replication
x <- sample(x=1:10,          # sample from 1, 2, 3, ..., 10
            size=10,         # sample of 10
            replace = TRUE)  # allowed to sample the same item multiple times
```

**Solution**: To obtain the third item of the vector, x, we use brackets and the index number 3.

```
(third.item = x[3])
```

```
[1] 5
```

### 1.2 Part b

**Solution**: Below, I create a data frame filled with random observations.

```
set.seed(7272) # sets the randomization seed for replication
df <- data.frame(x1 = sample(x=1:10,          # sample from 1, 2, 3, ..., 10
                             size=10,         # sample of 10
                             replace = TRUE), # allowed to sample the same item multiple times
                 x2 = sample(x=1:10,          # sample from 1, 2, 3, ..., 10
                             size=10,         # sample of 10
                             replace = TRUE), # allowed to sample the same item multiple times
                 x3 = sample(x=1:10,          # sample from 1, 2, 3, ..., 10
                             size=10,         # sample of 10
                             replace = TRUE)  # allowed to sample the same item multiple times,
)
```

### 1.3 Part b

Use the `head(...)` function to peek at the data frame.

**Solution**: The `head(...)` function allows us to view the top 6 rows (by default) of our data frame.

```
(sneak.peek = head(df))
```

```
  x1 x2 x3
1  8  1  8
2  3  7  2
3  5  3  3
4  3  6  2
5  2  2  3
6  3  8 10
```

## 1.4 Part c

Ask for the column `x1`.

**Solution**: We can view the column `x1` by using the `$` operator.

```
(col.x1 = df$x1)
```

```
 [1] 8 3 5 3 2 3 1 2 6 2
```

## 1.5 Part d

Ask for the fifth row of the data frame.

**Solution**: We can view the fifth row of the df by using brackets. The syntax for pulling data with brackets is [rows, cols], where leaving an entry blank tells `R` to pull *all* rows, or *all* columns (depending on which is left blank). Because we want to view all columns of row 5, we leave the columns slot blank.

```
(row.5 = df[5, ])
```

```
  x1 x2 x3
5  2  2  3
```

## 1.6 Part e

Ask for the the value of `x1` in the fifth row of the data frame.

**Solution**: To obtain the value of `x1` in the fifth row of the data frame, we again use brackets. However, this time, we specify that we want the first column, as *part d* has shown as that `x1` is the first column of our df.

```
(part.e = df[5,1])
```

```
[1] 2
```

## 1.7 Part f

Ask for the the value of in the third column and the fifth row of the data frame.

**Solution**: To obtain the value in third column and the fifth row of the data frame, we follow a similar procedure to *part e*.

```
(part.f = df[5,3])
```

```
[1] 3
```

## 1.8 Part g

Create a sequence from 1 to 10 by 2 and use it to print the odd rows of the data frame.

**Solution**: To create such a sequence, we use the `seq` function. The syntax of the function in this instance is intuitive, using `from`, `to`, and `by` as arguments. We can then use this list of odd numbers to pull the odd row numbers of our df by using brackets with the generated vector as our row indices.

```
(odd.indices = seq(from = 1, to = 10, by = 2))
```

```
[1] 1 3 5 7 9
```

```
(odd.rows = df[odd.indices,])
```

```
  x1 x2 x3
1  8  1  8
3  5  3  3
5  2  2  3
7  1 10  1
9  6  2 10
```

## 1.9 Part h

**Solution**: Below, I create an empty column called `x12`. Fill in the details of the `for(...)` loop to ensure `x12` is the product of `x1` and `x2`.

```r
n <- nrow(df)                # How many rows to we have to fill?
df$x12 <- rep(NA, nrow(df))   # Create an empty column for x12
for(i in 1:nrow(df)){
  (df[i,4] = df[i,1]*df[i,2])
}
df
```

```
   x1 x2 x3 x12
1   8  1  8   8
2   3  7  2  21
3   5  3  3  15
4   3  6  2  18
5   2  2  3   4
6   3  8 10  24
7   1 10  1  10
8   2  6  1  12
9   6  2 10  12
10  2  4  1   8
```

We know that x12 will be the fourth column, so we loop over each row in the fourth column and multiply the value of x1 and x2 for each row `i`.

## 1.10 Part i

Write a function called `calculate.score(...)` that takes three arguments representing `x1`, `x2`, and `x3`) and returns a single value based on the formula:

$$Score = (x_1 \times 2) + x_2 - x_3$$

Use your function to create a new column `total.score` in our data frame `df`.

**Solution**: We use the $ operator and our function to generate such a column.

```r
calculate.score = function(x1,x2,x3){
  score = (x1 * 2) + x2 - x3
}
df$total.score = calculate.score(df$x1,df$x2,df$x3)
df
```

```
   x1 x2 x3 x12 total.score
1   8  1  8   8           9
2   3  7  2  21          11
3   5  3  3  15          10
4   3  6  2  18          10
5   2  2  3   4           3
6   3  8 10  24           4
7   1 10  1  10          11
8   2  6  1  12           9
9   6  2 10  12           4
10  2  4  1   8           7
```

## 1.11 Part j

Create a function called `evaluate.row(...)` that takes one argument and returns "low" when the argument is less than 4, "mid" when the argument is between 4 and 7 (inclusive), and "high" when the arguement is 8 or larger. Then, use `sapply(...)` to apply it to column `x1`.

**Solution**:

```r
evaluate.row = function(x){
  if(x<4){
    "low"
  }else if(x<=7){
    "mid"
  }else{
    "high"
  }
}

part.j = sapply(df$x1,evaluate.row)
```

## 1.12 Part k

Did we need to use loops or functions in (h.)-(j.)? That is, can we use vectorization to attain the same results in 1 line each? Where it is possible, write the line of code. Where it is not, explain why.

**Solution**: We did not need to use loops for part *(h)*. We could have constructed the vector using the code below:

```r
df$hypothetical.h = df$x1 * df$x2
```

We could have completed part *(i)* without a function in a similar manner:

```r
df$hypothetical.i = (df$x1*2) + df$x2 - df$x3
df
```

```
   x1 x2 x3 x12 total.score hypothetical.h hypothetical.i
1   8  1  8   8           9              8              9
2   3  7  2  21          11             21             11
3   5  3  3  15          10             15             10
4   3  6  2  18          10             18             10
5   2  2  3   4           3              4              3
6   3  8 10  24           4             24              4
7   1 10  1  10          11             10             11
8   2  6  1  12           9             12              9
9   6  2 10  12           4             12              4
10  2  4  1   8           7              8              7
```

Part *(j)* was completed without a loop; however, it is possible to complete a similar task with a loop:

```r
df$category = rep(NA,nrow(df))
for(i in 1:nrow(df)){
if(df$x1[i]<4){
    df$category[i] = "low"
  }else if(df$x1[i]<=7){
    df$category[i] = "mid"
  }else{
    df$category[i] = "high"
  }
}
```

We could also avoid the block structure and loop structure entirely for part *(j)*, and use the vectorized form of `if` (though it's definitely a bit more cluttered):

```r
df$category = ifelse(df$x1<4,"low",
                     ifelse(df$x1<=7,"mid",
                     "high"))
```

# 2 Complete Tasks for One Recipe

## 2.1 Part a

Install and load the `rvest` package (Wickham 2025).

**Solution**: NOTE: Downloaded library installed in the console using `install.packages("...")`.

```r
library("rvest")
```

## 2.2 Part b

Load the html of the `website/KimchiGrilledCheese.html` using the `read_html()` function and save the result to an object called `recipe.item`. We will use the Kimchi Grilled Cheese recipe as our prototype and extend this workflow to all recipes, so try to be as general as possible. If you do look at it, you'll notice it contains the html we saw in the developer tools in class. **Hint**: You can use `read_html(...)` like `read.csv(...)`. Don't forget you can use the documentation to help use it.

**Solution**:

```r
recipe.item = read_html("website/KimchiGrilledCheese.html")
```

## 2.3 Part c

Open the html file in a web browser and open developer tools. In chrome-based browsers, you can do this by pressing the three verticle dots in the upper-right corner, clicking "more tools", then "developer tools".

Find the name of the Ingredients section of the website and pull the HTML of the function using `html_element(...)` and save the results to an object called `ingredients.section`. **Hint**: You can ask for elements by id using a preceeding "#". See `?html_element(...)` for a helpful example.

**Solution**: The name of the ID is `ingredients`.

```r
(ingredients.section = html_element(recipe.item,"#ingredients"))
```

```
{html_node}
<section id="ingredients" class="level2">
[1] <h2 class="anchored" data-anchor-id="ingredients">Ingredients</h2>
[2] <ul>\n<li>1/8 tsp chili flakes</li>\n<li>4 garlic cloves</li>\n<li>5 oz s ...
```

## 2.4 Part d

Now, we want to obtain all of the itemized items. Find the element type the individual ingredients and pull all of them from `ingredients.section` using `html_elements(...)` (note the added s) and save the results to an object called `ingredients`. **Hint**: You can ask for elements by type by simply specifying the tag (e.g., "p" for paragraph). See '?html_element for a helpful example.

**Solution**: We can see from part *(c)* that the ingredients are type `<li>`.

```r
(ingredients = html_elements(ingredients.section,"li"))
```

```
{xml_nodeset (9)}
[1] <li>1/8 tsp chili flakes</li>
[2] <li>4 garlic cloves</li>
[3] <li>5 oz spinach</li>
[4] <li>1 tsp gochujang</li>
[5] <li>3 tsp mayonnaise</li>
[6] <li>2/3 cup kimchi</li>
[7] <li>4 slices of bread</li>
[8] <li>2/3 cup cheddar cheese</li>
[9] <li>4 tablespoons everything seasoning.</li>
```

## 2.5 Part e

Similar to Part c. Find the name of the Instructions section of the website and pull the HTML of the function using `html_element(...)` and save the results to an object called `instructions.section`.

**Solution**: The name of the instructions section is `instructions`.

```r
instructions.section = html_element(recipe.item,"#instructions")
```

## 2.6 Part f

Now, we want to obtain all of the enumerated items. Find the element type the individual instructions and pull all of them from `instructions.section` using `html_elements(...)` (note the added s) and save the results to an object called `instructions`.

**Solution**: The instructions are similarly, type `<li>`.

```r
(instructions = html_elements(instructions.section,"li"))
```

```
{xml_nodeset (5)}
[1] <li>In a frying pan, heat 2 tablespoons olive oil with chili flakes.</li>
[2] <li>Add spinach and cook until wilted.</li>
[3] <li>Mix gochujang and mayonnaise in a small bowl and spread on one side o ...
[4] <li>Build sandwiches by layering spinach, kimchi, and cheese between two  ...
[5] <li>Fry in butter until golden brown on both sides.</li>
```

## 2.7 Part g

Find the class of the recipe image and pull the HTML using `html_element(...)` and save the results to an object called `image.element`. **Hint**: You can ask for elements by class using a preceeding ".". See `?html_element(...)` for a helpful example. Further, note that HTML elements may have more than one class (separated by a space). When that is the case, you need to choose one.

**Solution**: The image element is class `.img-fluid` and `figure-img`. I arbitrarily chose to specify the former.

```r
image.element = html_element(recipe.item,".img-fluid")
```

## 2.8 Part h

Use the `html_attr(...)` function to pull the source link ("src"). Then, use `paste(...)` to prepend the source link with "website/" so we have the full link. **Note**: This would be like adding "https://www.website.com" to get the

absolute link.

**Solution**:

```
(src = html_attr(image.element, "src"))
```

```
[1] "images/KimchiGrilledCheese.jpg"
```

```
(image.url = paste("website/",src, sep = ""))
```

```
[1] "website/images/KimchiGrilledCheese.jpg"
```

## 2.9 Part i

Now that we have all the things we need, let's try to print the recipe. Below, I have written code to print from the objects. One by one, remove `#| eval: false` from the YAML header and add `#| echo: false` and `#| results: 'asis'`, and test. Let me know if you're stuck!

**Image**

Figure 1: Kimchi Grilled Cheese

**Ingredients**

- 1/8 tsp chili flakes
- 4 garlic cloves
- 5 oz spinach
- 1 tsp gochujang
- 3 tsp mayonnaise
- 2/3 cup kimchi
- 4 slices of bread
- 2/3 cup cheddar cheese
- 4 tablespoons everything seasoning.

**Instructions**

1. In a frying pan, heat 2 tablespoons olive oil with chili flakes.
2. Add spinach and cook until wilted.
3. Mix gochujang and mayonnaise in a small bowl and spread on one side of each slice of bread. Sprinkle with everything seasoning and press it into the bread.

4. Build sandwiches by layering spinach, kimchi, and cheese between two slices of bread.
5. Fry in butter until golden brown on both sides.

# 3 Complete a Full Menu!

Open `Menu.qmd` and add code to complete the following.

1. Randomly select three dinner recipes and one breakfast recipe at random. **Hint:** Use the `sample(...)` function.
2. Pull the image, ingredients, and instructions for each selected recipe.
3. Combine all of the ingredients into a grocery list on the first page.
4. Print the image, ingredients, and instructions for each receipe on the subsequent pages.

When you render this document, no code should be visible. Instead, you should see a five-page document as described above.

# 4 Describe your work!

The website construction went well in the end; however, I ended up creating it with clunky code. I used several loops and repeated the scraping process twice (once for the grocery list, once for the recipes themselves). I also used a lot of manual spacing to accomodate my loop which potentially makes the code less dynamic. Overall, there are some redundancies in my code, but for someone with minimal coding experience/someone for whom coding does not come easily, I am somewhat pleased with the output.

## 4.1 Why a fake website?

Read https://www.scrapingbee.com/blog/is-web-scraping-legal/. Originally, I conceived doing this with recipes from Purple Carrot (my favorite subscription service). Being a large company, their terms of service are *very* long and precluded us from copying their recipes. We also checked a few smaller recipe websites we like and even they had terms of service that restricted automated collection of data.

## 4.2 Conditionals

Did you use conditional statements in your code? If yes, how. If not, are there places you could have used them but did something else?

**Response:** Yes, I used a conditional statement to check if the recipe was the last recipe which, if it was, would label it as the breakfast recipe. This worked because of how I created the object `all.meals`, with breakfast concatenated to the last entry.

## 4.3 Loops

Did you use loops in your code? If yes, how. If not, are there places you could have used them but did something else?

**Response:** Yes, I used two for loops. The first of which I used to build the grocery list, the second I used to print each recipe's image, ingredients, and instructions.

## 4.4 Functions

Did you use functions in your code? If yes, how. If not, are there places you could have used them but did something else?

**Response:** I did not create any functions of my own. In hindsight, it would've been efficient to create a function that could take html as the input and print the associated image, ingredient, and instructions combo.

## References

Wickham, Hadley. 2025. *Rvest: Easily Harvest (Scrape) Web Pages*. https://doi.org/10.32614/CRAN.package.rvest.